Name:	A.M.T DARSHANA ABEYSINGHE
Index Number:	C209943
NIC:	200524201565

Student Registration System Of Stevens University

Table of Contents

1. Abstraction / Introduction

This project is a dynamic web application designed to streamline the process of managing student information within a private educational institution, namely Stevens Institute of Technology. The application focuses on automating tasks like student registration, updating personal information, deleting outdated records, and searching for students based on their NIC. A secure login system is also included to restrict access to authorized personnel only.

The development process utilized the XAMPP server environment, integrating core web technologies such as HTML and CSS for the user interface, PHP for server-side logic, and MySQL as the backend database. The project was designed with scalability and ease-of-use in mind, ensuring that even non-technical staff can operate the system effectively.

The overall goal is to minimize manual effort, improve data accuracy, and enhance the user experience by offering a clean, responsive interface and fast data processing. All design decisions were based on standard web development principles and practical usability.

2. Objective of the Project

- To build a fully functional web-based student management system tailored for a private educational institution.
- To develop a secure and easy-to-navigate user interface that simplifies student data handling for admins.
- To incorporate CRUD operations (Create, Read, Update, Delete) for managing student records.
- To implement a login system to restrict unauthorized access to sensitive student data.
- To offer an intuitive course listing and registration module that helps guide new students.
- To ensure that the backend system stores information reliably using a MySQL relational database.

3. Tools and Technologies Used

Technology	***Description***
HTML	Used to create the structure of each webpage, including forms and content sections.
CSS	Styled the pages to enhance visual appearance and ensure responsive layouts.
PHP	Handled form submissions, backend logic, and database queries.
MySQL	Stored student data, including personal details and course enrollments.
XAMPP	Used as the local development environment for integrating PHP and MySQL.
JavaScript	Provided minor enhancements in form validation and interactivity.

4. Project UI and Explanation

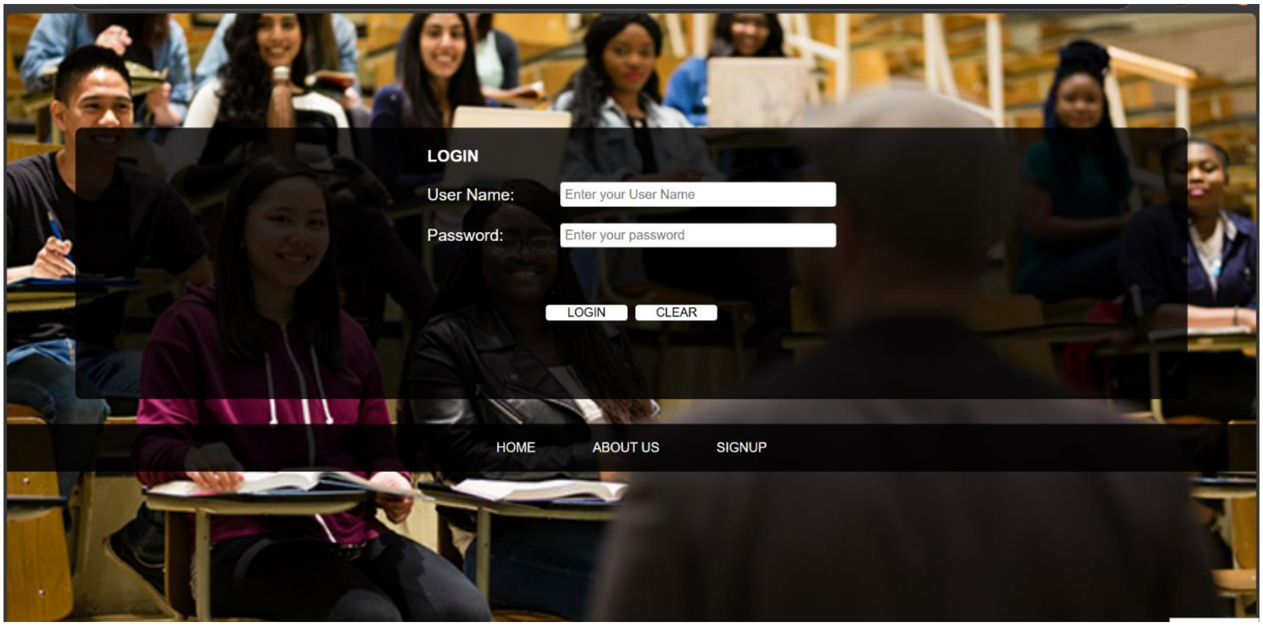
A. Home Page (Home.html)

The dashboard that users land on post-login. It includes:
- A background image with an institutional logo.
- A navigation bar that provides links to other key parts of the website.
- Sticky menu to ensure users can navigate easily even when scrolling.

B. About Page (About_us.html)

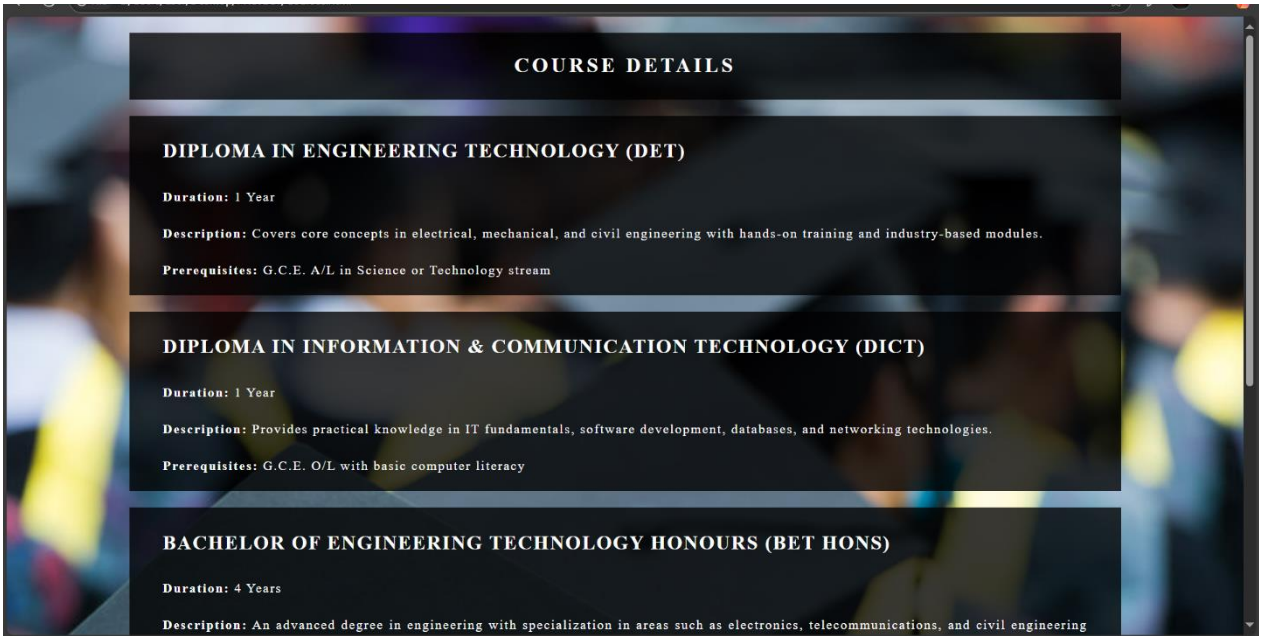
Presents the mission and background of Stevens Institute of Technology.
- Describes the institution's foundation, focus on innovation, and public responsibility.
- Offers insight into how the institution aims to shape civic-minded graduates.
- Design uses center-aligned text with a focus on readability and aesthetics.

C. Login Page (Login.html and login.php)



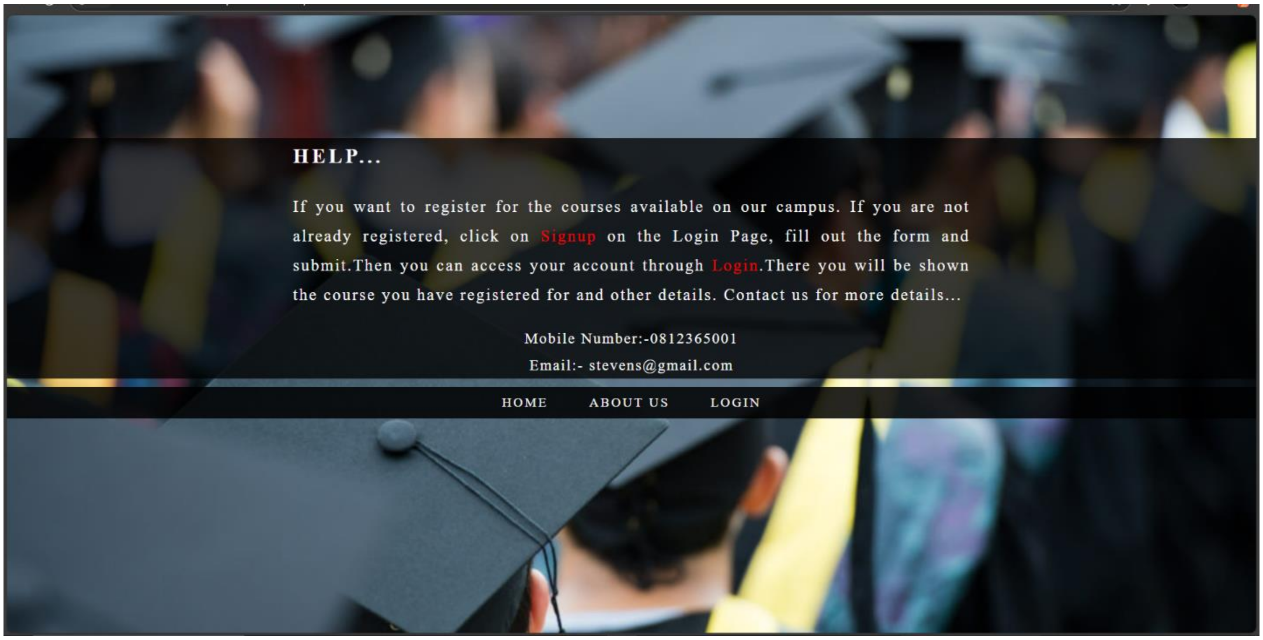
This is the secure entry point of the system, designed for administrative users.
- Contains two fields: username and password.
- Backend logic (login.php) checks credentials against predefined values.
- If authentication succeeds, the user is redirected to the Home page.
- Invalid credentials prompt an alert and reload the login screen.

D. Courses Page (Courses.html)



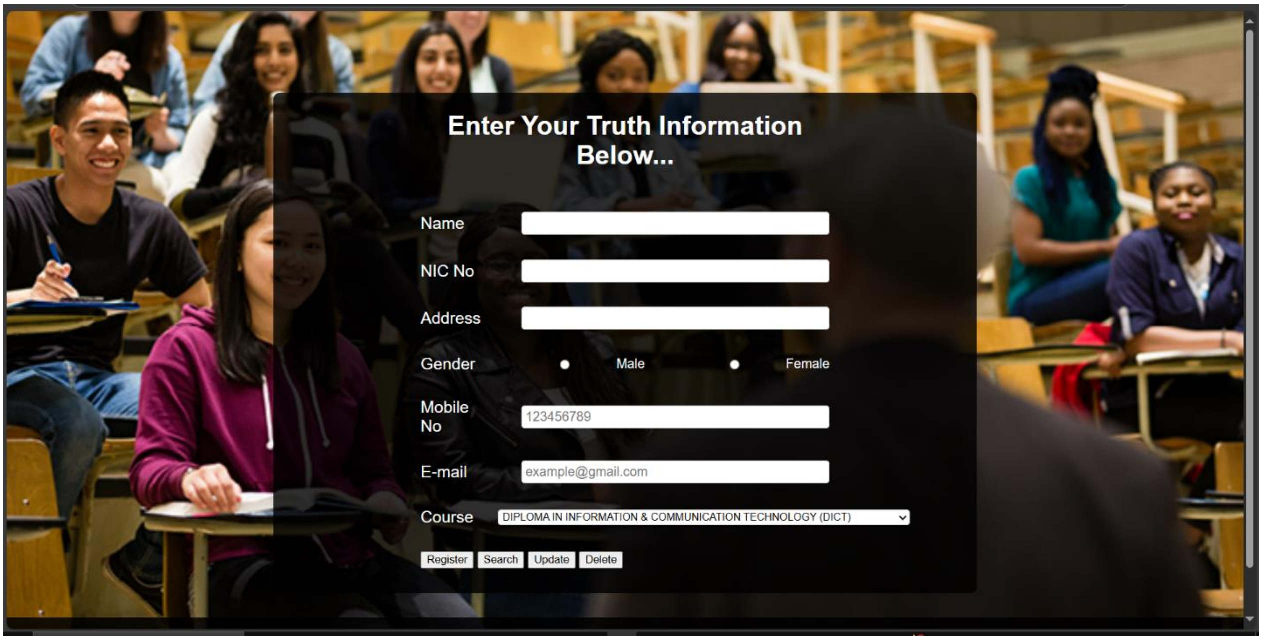
Displays the list of all available programs offered at the institution.
- Programs include both diploma and bachelor's level courses.
- Each course section describes the duration, prerequisites, and a summary.
- Background image and transparent content panel enhance visual design.

E. Help Page (Help.html)



Provides step-by-step guidance for students who need assistance with registration and login.
- Instructions include how to use the Signup and Login functionality.
- Contact information such as mobile number and email is prominently displayed.
- Styled consistently to match the rest of the website with dark overlays.

F. Signup Page (Signup.html / Register.php / Search.php)



The student registration form allows new students to input their personal and academic details.
- Fields include name, NIC, address, gender, mobile number, email, and selected course.
- Four operations supported via buttons: Register, Search, Update, and Delete.
- The form sends data to Register.php where actions are handled with PHP and MySQL queries.
- Validation is done via required fields and controlled select options.

5. Database Design

Database Name	Table Name	Fields
Student_Info	Student_Details	name nic address gender mobile email course

Primary key is NIC validated to avoid duplicates(Signup Page).

6. Challenges Faced and Solution

Challenge	Solution
Database connection failure	Resolved by checking server name, username, and empty password in XAMPP settings.
Incorrect form submissions	Added "required" attribute in form inputs and validated form names.
Page navigation not working	Fixed by correcting file paths and using relative linking in the menu.
Unresponsive layout	Enhanced CSS by using flexbox and consistent padding/margin structure.
Data not saving in DB	Matched HTML form field names with PHP variables and SQL column names.

7. Submit Web Application Development Project

https://drive.google.com/drive/folders/1gtvLAD3-7F8qXDKKJC5HqcS0hbfyB9zp?usp=drive_link